

● RAG Learning

B2.5 How can lifestyle, genes and the environment affect health?

OCR B GCSE Biology (J257) Twenty First Century Science — Topic B2: Keeping healthy

1. Risk factors

A **risk factor** is a feature of a person's lifestyle, environment or genome that increases their probability of developing a particular disease.

Risk factor	Diseases linked to it
Smoking	Lung cancer, COPD, cardiovascular disease, low-birth-weight babies.
Excessive alcohol	Cirrhosis, liver cancer, brain damage, fetal alcohol syndrome.
Poor diet (high fat / sugar / salt)	Obesity, type 2 diabetes, CVD, hypertension.
Lack of exercise	Obesity, CVD, type 2 diabetes, depression.
Obesity (BMI > 30)	Type 2 diabetes, CVD, osteoarthritis, some cancers.
UV radiation / sunlight	Skin cancer (melanoma).
Carcinogens (e.g. asbestos, tobacco tar)	Cancers (mesothelioma, lung).
Genetic factors	BRCA1/2 → breast & ovarian cancer; familial hypercholesterolaemia → CVD.

2. Cardiovascular disease (CVD)

The leading cause of death in the UK. Includes coronary heart disease (blocked coronary arteries), heart attack, stroke and heart failure. Caused mainly by atherosclerosis — fatty plaque builds up in arteries, narrows them, restricts blood flow.

CVD risk factor	Why it increases risk
High saturated fat / cholesterol intake	More LDL cholesterol → fatty plaque formation in arteries.
High salt intake	Raises blood pressure → damages artery walls.
Smoking	Nicotine raises BP; CO reduces oxygen carrying.
Inactivity	Weight gain, insulin resistance, raised cholesterol.
Genetics (family history)	Inherited high cholesterol or BP.

3. Cancer

Cancer is a group of diseases caused by uncontrolled cell division producing a tumour.

Benign tumours stay in one place; **malignant** tumours invade neighbouring tissue and can spread (metastasis) via blood or lymph.

Cancer	Lifestyle/environmental factor
Lung cancer	Smoking, radon, air pollution.
Skin cancer (melanoma)	UV exposure, fair skin (genetic).
Liver cancer	Alcohol, hepatitis B/C virus, aflatoxins.
Breast cancer	Genetics (BRCA1/2), age, hormonal factors.

Bowel cancer	Low fibre, processed meat, lack of exercise.
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4. Mental health

Mental health is part of overall health. **Depression**, **anxiety** and **schizophrenia** are influenced by genes, life events, brain chemistry and lifestyle (sleep, exercise, social contact). Long-term physical illness raises risk; conversely, depression worsens recovery from physical disease.

5. Sampling and correlation

Population health data is gathered by surveys and registries. A **positive correlation** between two factors (e.g. smoking and lung cancer) does not by itself prove that one causes the other, but a strong, dose-dependent and biologically plausible link is good evidence.

Exam Tip: Use the Bradford Hill style "evidence" terms when evaluating data: dose–response, consistency across studies, biological plausibility, temporality (cause precedes effect).

6. Summary checklist

- Define risk factor and give one example for each: lifestyle, genetic, environmental.
- Explain the role of fatty plaque in CVD.
- Distinguish benign and malignant tumours.
- List three lifestyle factors associated with bowel cancer.
- Discuss links between mental and physical health.
- Explain why correlation does not prove causation in health data.